Lecture 1 - 18/10

Wednesday, October 21, 2026 8:55 PM

Intro+Four ierI_slides

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https://docs.google.com/presentation/d/1sQxKRff19xBYH1eBXH5CaX7X3XaHl620Ux-pl05r6tY/edit?slide=id.p5#slide=id.p5

Fourier

Introduction to Computer Vision

Michal Irani, Meirav Galun, Ronen Basri, Shai Bagon

Teaching Assistants

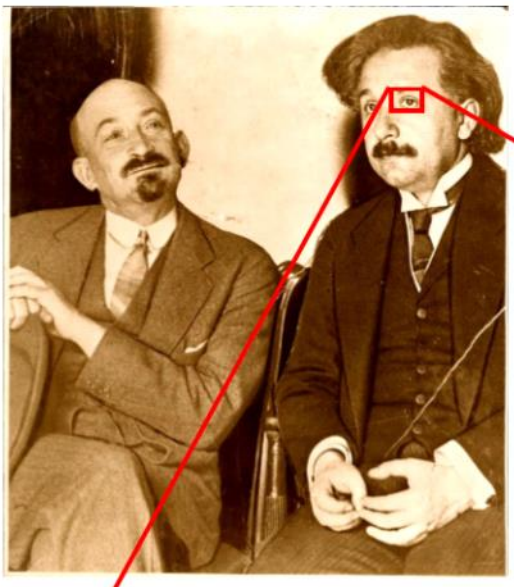
Yuval Golbari, Daniel Benaroya, Eliya Olkhovsky

A few announcements

- **Course website – on Moodle**
  (schedule, lesson slides, exercises, announcements, recordings, etc.)

- ***Vision&AI* Seminar:** (not for credit, but highly recommended)

  Thursdays at 12:15-13:15
  To be added to mailing list – *visit SEMINAR-SIGNUP page:*
  https://www.weizmann.ac.il/math/seminar-subscribe
  1st meeting this year – on Thursday Oct 22 (this week).



Digital Image

Pixels:

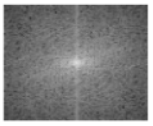
0 = Black

.
.
.

255 = White

137	74	52	16	128	217	207	221	220	179	188	193	195	189	188	193
73	64	23	68	228	243	225	120	94	138	140	116	210	173	162	142
29	59	43	246	246	151	99	74	185	188	214	205	127	138	203	186
65	19	187	244	170	37	153	255	233	245	255	236	252	182	123	197
30	72	255	118	20	232	235	218	184	50	41	9	55	147	207	110
19	139	108	42	244	253	130	75	5	207	40	73	31	81	11	181
42	147	69	235	187	81	222	236	59	62	4	55	0	141	81	9
33	111	231	139	67	217	255	240	20	119	155	158	39	91	84	15
76	251	160	71	195	255	241	255	255	55	54	80	176	188	245	231
255	155	103	237	224	240	255	253	250	255	248	243	247	227	194	137
237	239	255	222	220	219	205	191	203	206	180	168	147	140	96	85

Topics covered in the course



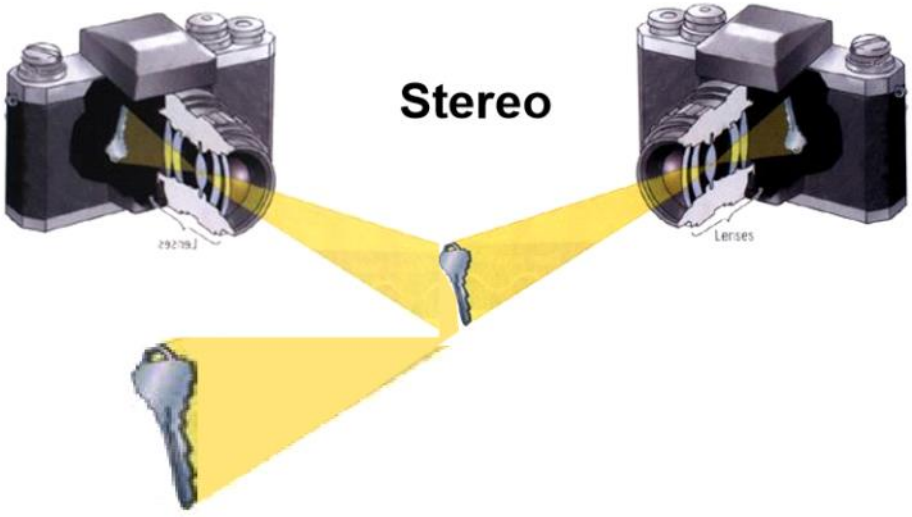
- **Fourier Transform & Applications - Michal** *(2 lessons)*
- **Geometry, Stereo, 3D Structure - Meirav** *(4 lessons)*
- **Motion & Video analysis – Michal** *(2 lessons)*
- **Lighting - Ronen** *(2 lessons)*
- **Patch-based methods – Shai** *(1 lesson)*
- **Boundaries & Segmentation – Shai** *(1 lesson)*

- **2 programming exercises (Python)** -- SUBMIT IN PAIRS
- **2 theoretical exercises** -- SUBMIT INDIVIDUALLY
- **EXAM**

- **2-point Practical Deep-Learning course (online with hands-on exercises)**
  given in parallel by Meirav Galun, Tali Dekel, Shai Bagon.
- **Both courses (*"Intro-to-Computer-Vision"* & *"DL course"*) are *prerequisites* for taking the *"ADLV course"* (Semester2)**



Structure from Motion



Panoramic Mosaic Image



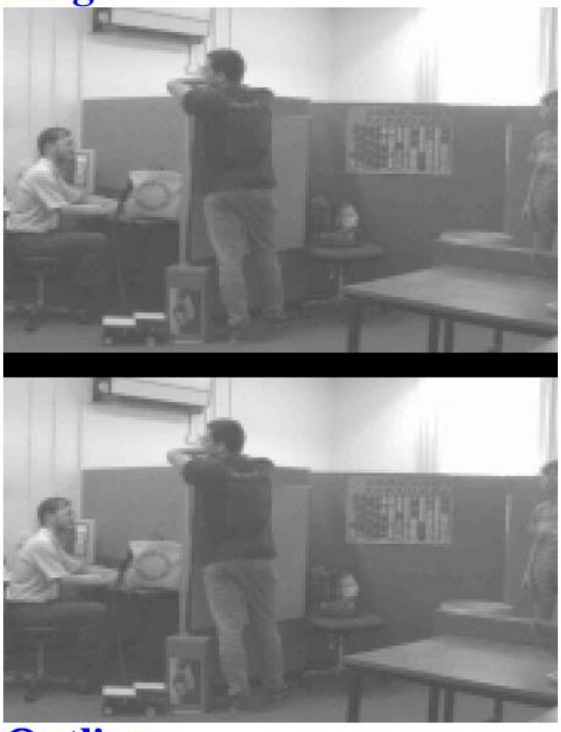
Original video clip

Generated Mosaic image



Video Removal

Original



Outliers

Original



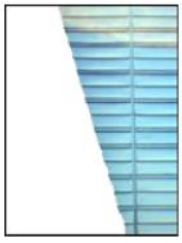
Synthesized

Segmentation and Boundaries












Learning ⇒ Usually requires LOTS of training data

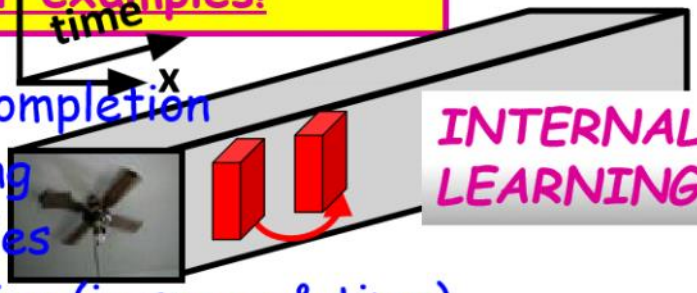
Exploit internal data redundancy (inside a single image or video)



True for most 5x5 patches in any natural image

l inference

nior examples!

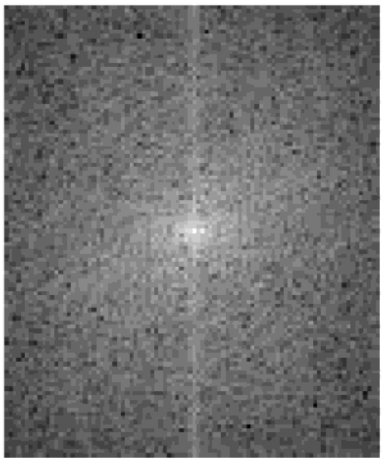


INTERNAL LEARNING

- Image/Video completion
- Image denoising
- Visual summaries
- Super-Resolution (in space & time)
- Image deblurring
- Image dehazing,
- Image compression …

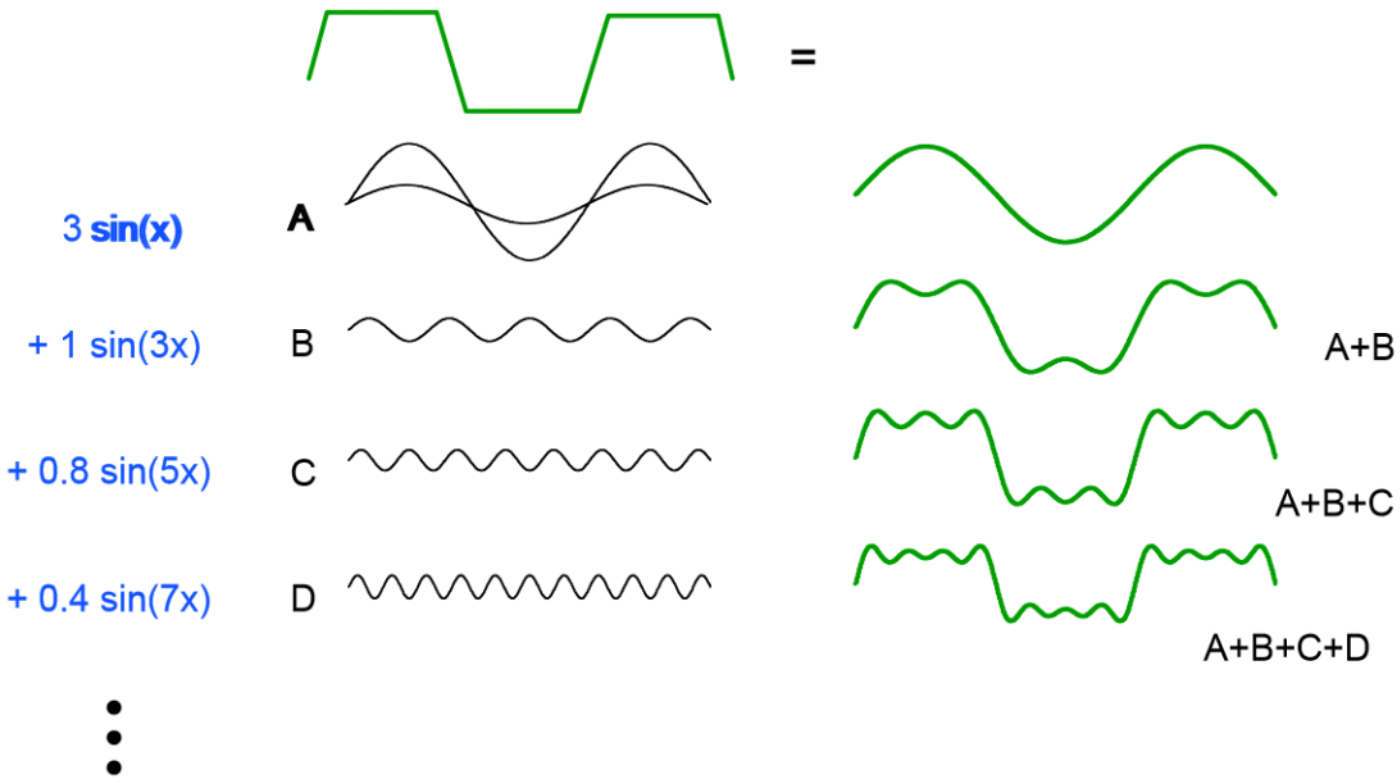
The Fourier Transform

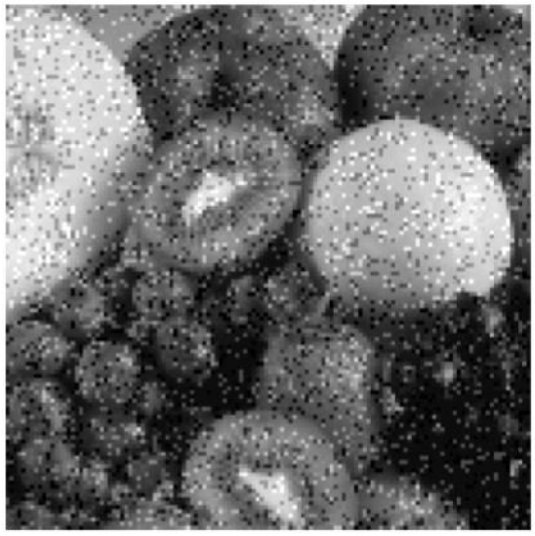


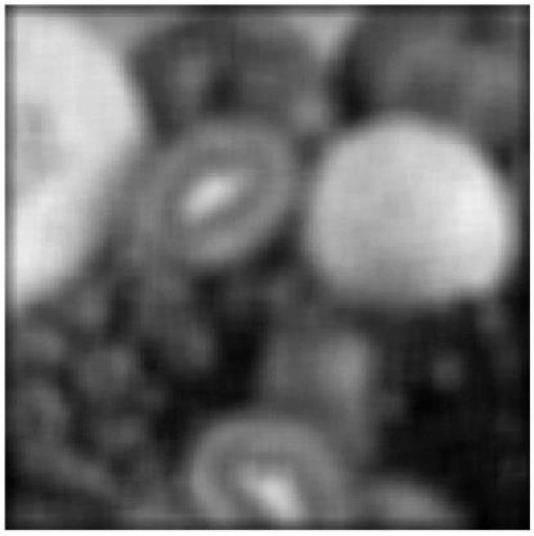


Jean Baptiste Joseph Fourier

A sum of sines and cosines







Higher frequencies due to sharp image variations (e.g., edges, noise, etc.)

Formulating the Fourier Transform

So how prominent is a frequency ***u*** in a function/image ***f*** ?

We will formulate it for:

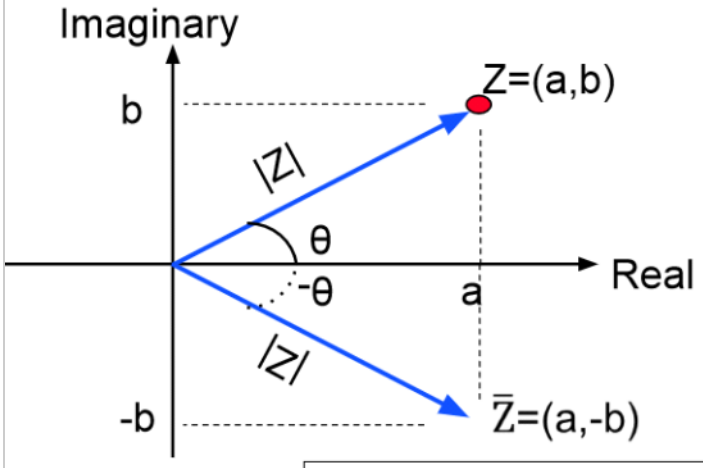
- The Continuous & Discrete functions ***f*** .
- 1D and 2D case.

The **Continuous** Fourier Transform

Basis functions:

$$\mathbf{e}^{2\pi iux} = \cos(2\pi ux) + i\sin(2\pi ux)$$

Complex Numbers



$$Z = \mathrm{Re}(Z) + i\,\mathrm{Im}(Z)$$
$$= a + ib$$
$$= |Z| e^{i\theta}$$

$$i = \sqrt{-1} \qquad (i^2 = -1)$$

$$e^{i\theta} = \cos\theta + i\sin\theta \qquad \text{(a unit vector)}$$

$$|Z| = \sqrt{a^2 + b^2} \qquad \text{(Magnitude)}$$

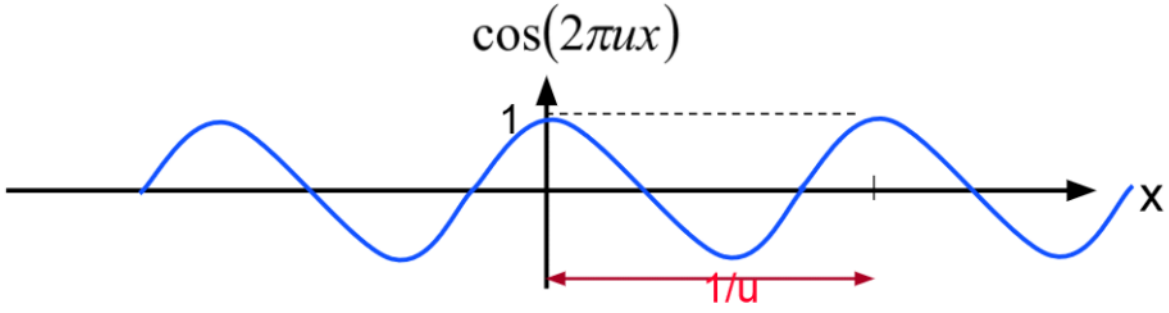
$$\theta = tg^{-1}(b/a) \qquad \text{(phase)}$$

$$\overline{Z} = Z^* = a - ib = |Z| e^{-i\theta} \qquad \text{(conjugate)}$$

The 1D Basis Functions $e^{2\pi iux}$

$$\mathbf{e}^{2\pi iux} = \cos(2\pi ux) + i\sin(2\pi ux)$$

$$\mathrm{Re}(e^{2\pi iux}) = \cos(2\pi ux)$$
$$\mathrm{Im}(e^{2\pi iux}) = \sin(2\pi ux)$$



– The wavelength is 1/u .

– The frequency is u .

The Continuous Fourier Transform

1D Continuous Fourier Transform:

$$f(x) = \int_{u=-\infty}^{\infty} \mathrm{F(u)} e^{+2\pi i u x} du = \Im^{-1}(F(u))$$

The Inverse Fourier Transform

$$F(u) = \int_{x=-\infty}^{\infty} \mathrm{f(x)} e^{-2\pi i u x} dx = \Im(f(x))$$

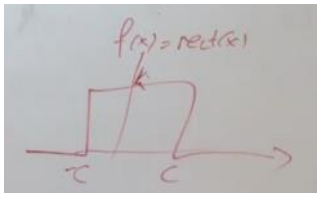
The Fourier Transform

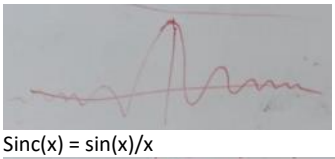
An orthonormal basis

$$F(u) = \int_{x=-\infty}^{\infty} \mathrm{f(x)} e^{-2\pi i u x} dx$$

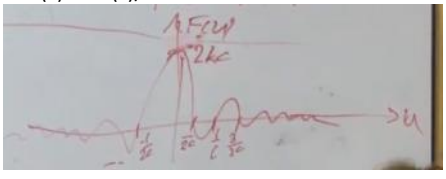
Some Fourier Transforms

Function	Fourier Transform
$f(x) = \text{rect}(x) = \begin{cases} k & -c \le x \le c \\ 0 & otherwise \end{cases}$	$F(u) = 2kc \cdot \text{sinc}(2\pi cu)$
$f(x) = \delta(x)$	$F(u) \equiv 1$
$f(x) =$	$F(u) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i u x} dx$
$f(x) =$	$= (e^{i\alpha x} + e^{-i\alpha x}$
$f(x) =$	





Sinc(x) = sin(x)/x



Gaussia

The Continuous Fourier Transform

1D Continuous Fourier Transform:

$$f(x) = \int_{u=-\infty}^{\infty} \mathrm{F(u)} e^{+2\pi i u x} du = \Im^{-1}(F(u))$$

The Inverse Fourier Transform

$$F(u) = \int_{x=-\infty}^{\infty} \mathrm{f(x)} e^{-2\pi i u x} dx = \Im(f(x))$$

The Fourier Transform

2D Continuous Fourier Transform:

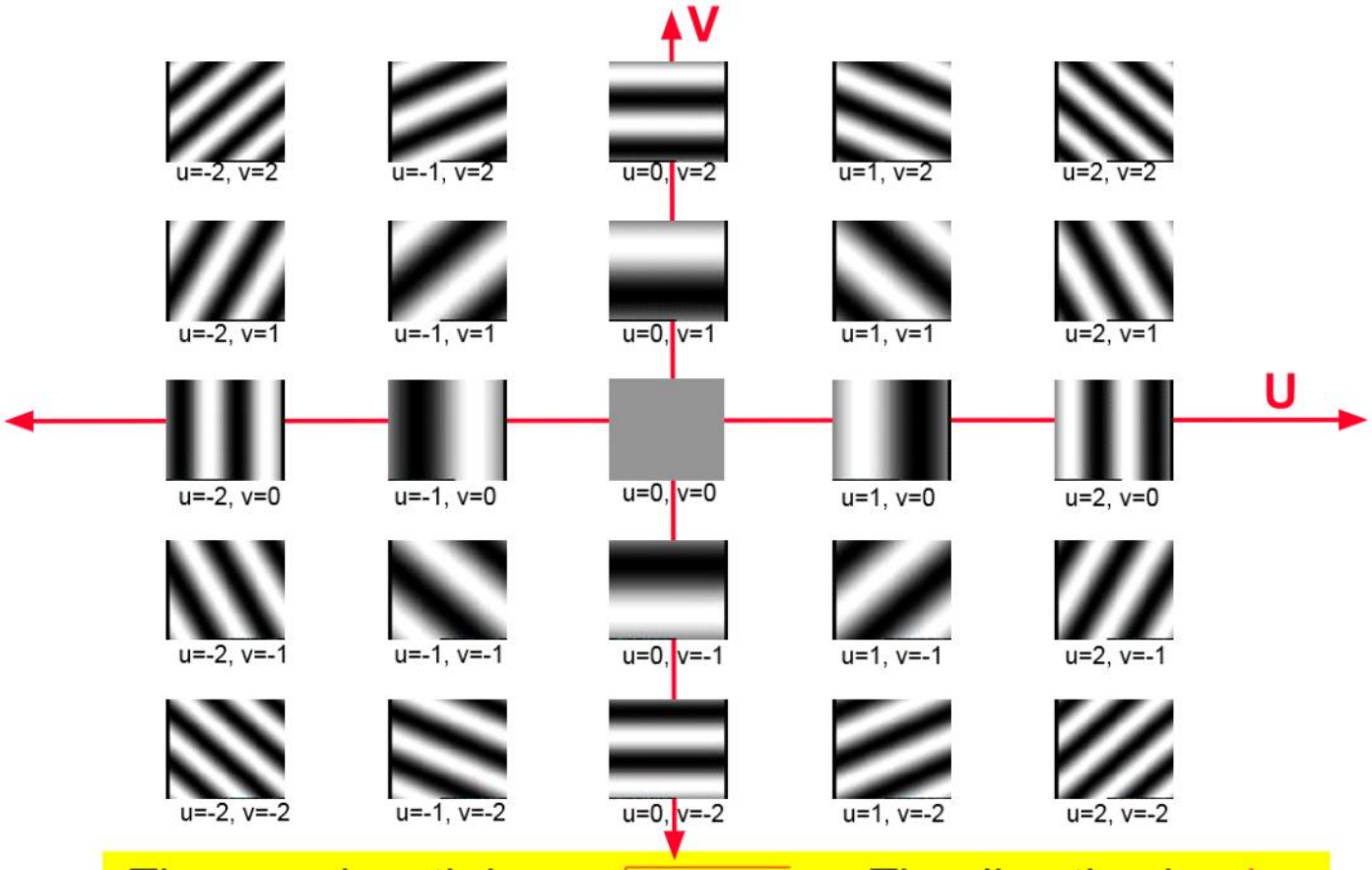
$$f(x,y) = \int_{u=-\infty}^{\infty} \int_{v=-\infty}^{\infty} \mathrm{F(u,v)} e^{+2\pi i (ux+\mathrm{vy})} du dv$$

The Inverse Transform

$$F(u,v) = \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} \mathrm{f(x,y)} e^{-2\pi i (ux+\mathrm{vy})} dx dy$$

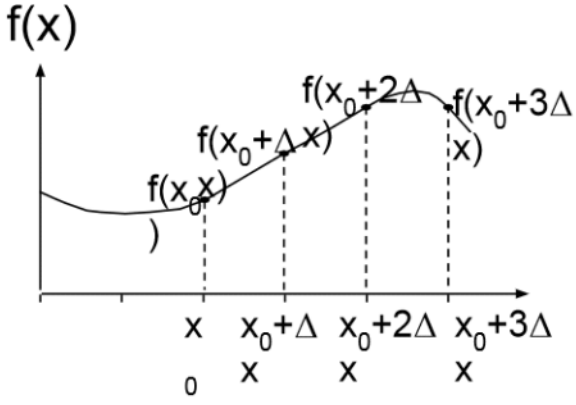
The Transform

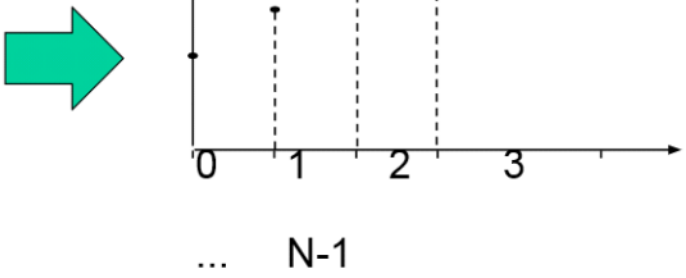
The 2D Basis Functions $e^{2\pi i(ux+vy)}$



The wavelength is $1/\sqrt{u^2+v^2}$. The direction is u/v .

Discrete Functions





The discrete 1D function f:

{ f(0), f(1), f(2), … , f(N-1) }

The Finite Discrete Fourier Transform

1D Finite Discrete Fourier Transform:

$$F(u) = \frac{1}{N}\sum_{x=0}^{N-1} f(x)e^{\frac{-2\pi iux}{N}}$$

(u = 0,..., N-1)

$$f(x) = \sum_{u=0}^{N-1} F(u)e^{\frac{+2\pi iux}{N}}$$

(x = 0,..., N-1)

2D Finite Discrete Fourier Transform:

$$F(u,v) = \frac{1}{N}\frac{1}{M}\sum_{x=0}^{N-1}\sum_{y=0}^{M-1} f(x,y)e^{-2\pi i(\frac{ux}{N}+\frac{vy}{M})}$$

(u = 0,..., N-1; v = 0,...,M-1)

$$f(x,y) = \sum_{u=0}^{N-1}\sum_{v=0}^{M-1} F(u,v)e^{+2\pi i(\frac{ux}{N}+\frac{vy}{M})}$$

(x = 0,..., N-1; y = 0,...,M-1)

About the Discrete Transform

1. f_d is a discrete sampling of the continuous $f(x)$ at gaps Δx

   ⇕

   F_d is a discrete sampling of the continuous $F(u)$ at gaps $\Delta u = \frac{1}{N\Delta x}$

$$\begin{array}{c} f(x) \leftrightarrow F(u) \\ \downarrow \Delta x \qquad \downarrow \Delta u = \frac{1}{N\Delta x} \\ f_d(i) \leftrightarrow F_d(j) \end{array}$$

2. **Periodicity of the discrete transform (both $f(x)$ and $F(u)$):**

   $F(u+N) = F(u)$

   $f(x+N) = f(x)$

$$F(u+N) = \frac{1}{N}\sum_0^{N-1} f(x)\, e^{-\frac{2\pi i x(u+N)}{N}} = \frac{1}{N}\sum_0^{N-1} f(x)\, e^{-\frac{2\pi i x u}{N}} \underbrace{e^{-2\pi i x}}_{=1} = F(u)$$

   $f(x+N) = f(x)$ *can be shown using the inverse transform*

3. **Computational complexity:** $O(N \cdot \log N)$
   (with FFT – the Fast Fourier Transform)

The Fourier Image

Image f



Fourier spectrum (magnitude)
log(|F(u,v)|)

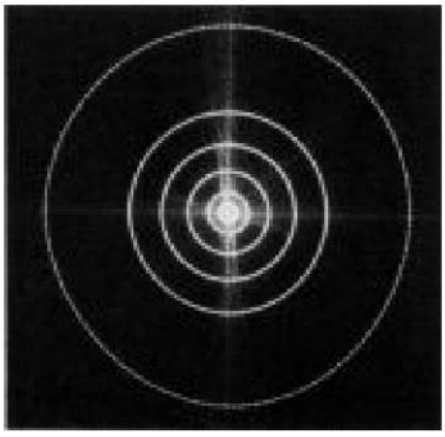


Frequency Bands

Image



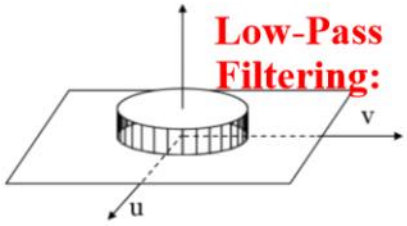
Fourier Spectrum



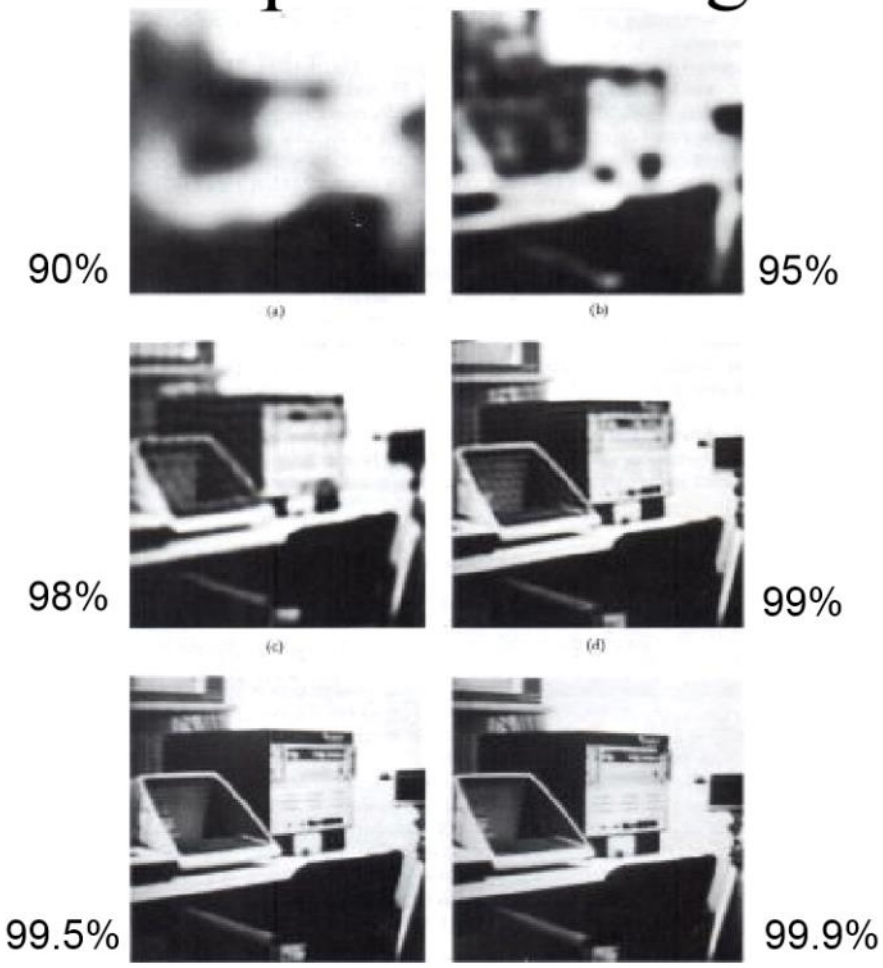
Percentage of image power enclosed in circles (small to large) :

90%, 95%, 98%, 99%, 99.5%, 99.9%

Low-Pass Filtering:



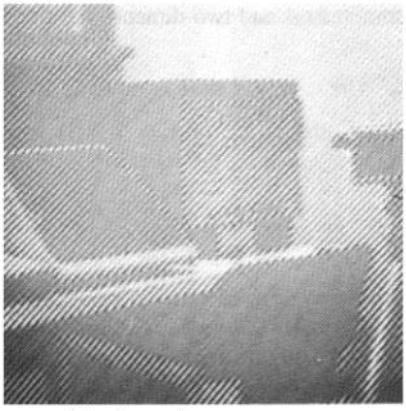
Low pass Filtering



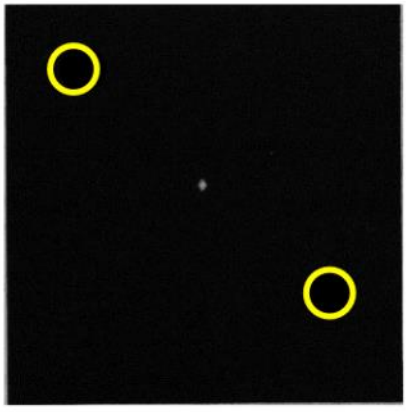




Noise Removal

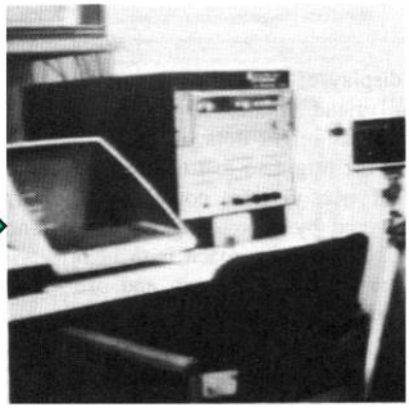


Noisy image



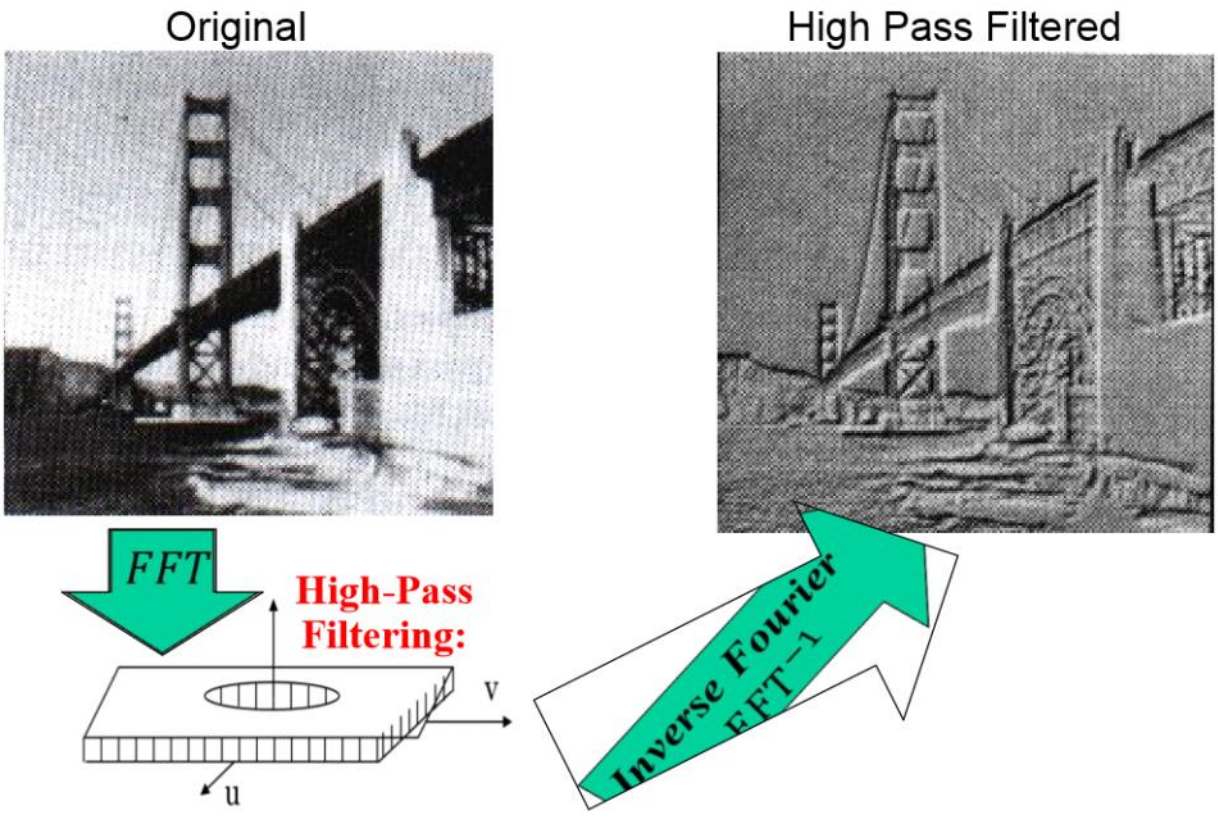
Fourier Spectrum (magnitude)

Inverse Fourier
(FFT^{-1})



Noise-cleaned image

High Pass Filtering



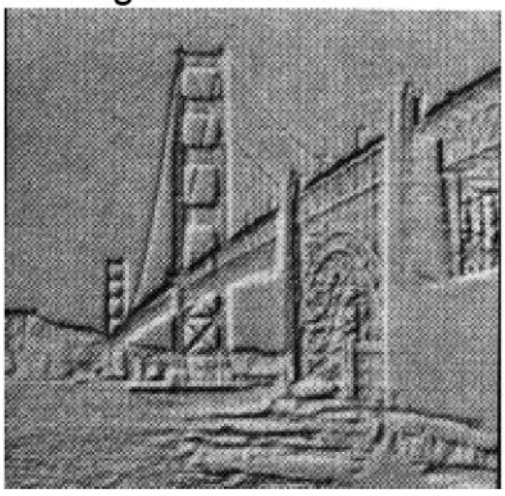
High Frequency Emphasis

Original



+

High Pass Filtered



High Frequency Emphasis

Original



High Frequency Emphasis



High Frequency Emphasis

Original



High Frequency Emphasis





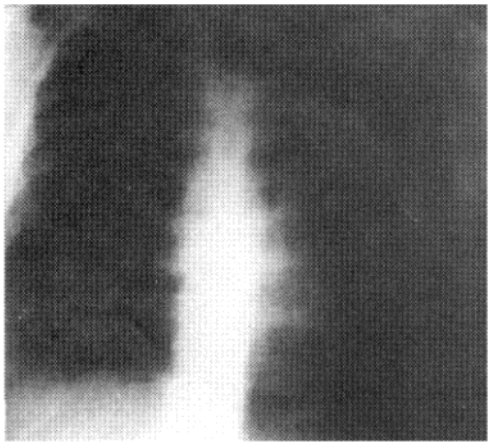
Original



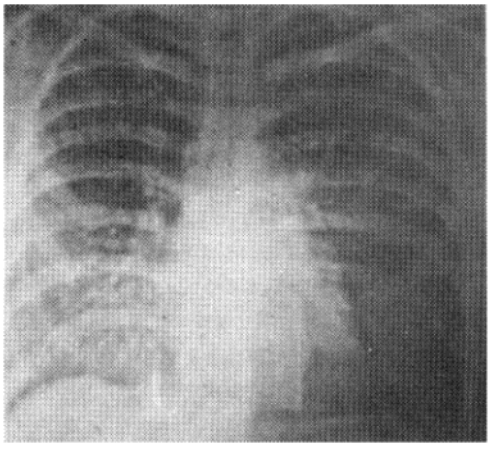
High Frequency Emphasis

High Frequency Emphasis

Original

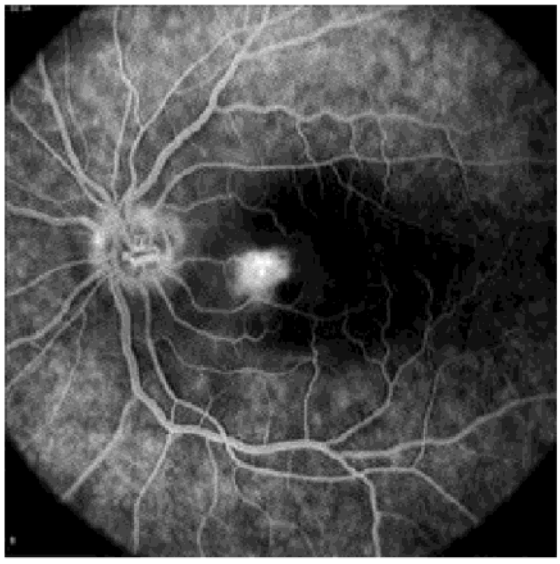


High Frequency Emphasis

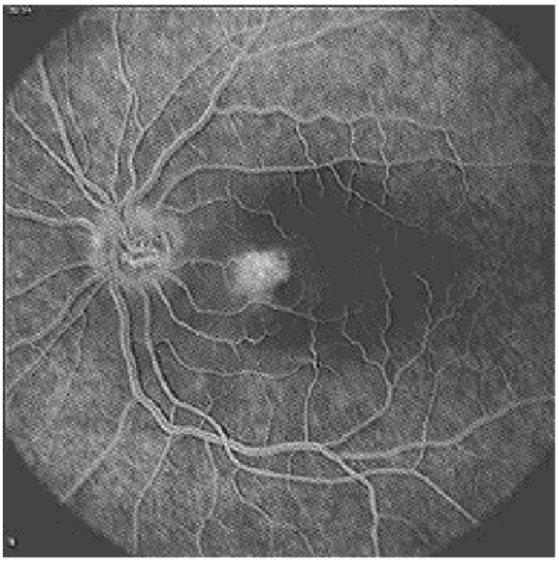


High Frequency Emphasis

Original



High Frequency Emphasis



Importance of Phase vs. Magnitude

- **Curious fact:**

  All natural images have approximately the same Fourier magnitude; but images look very different from one another…
  How come….?

  ➔ phase seems to matter more than magnitude.

- **Demonstration:**

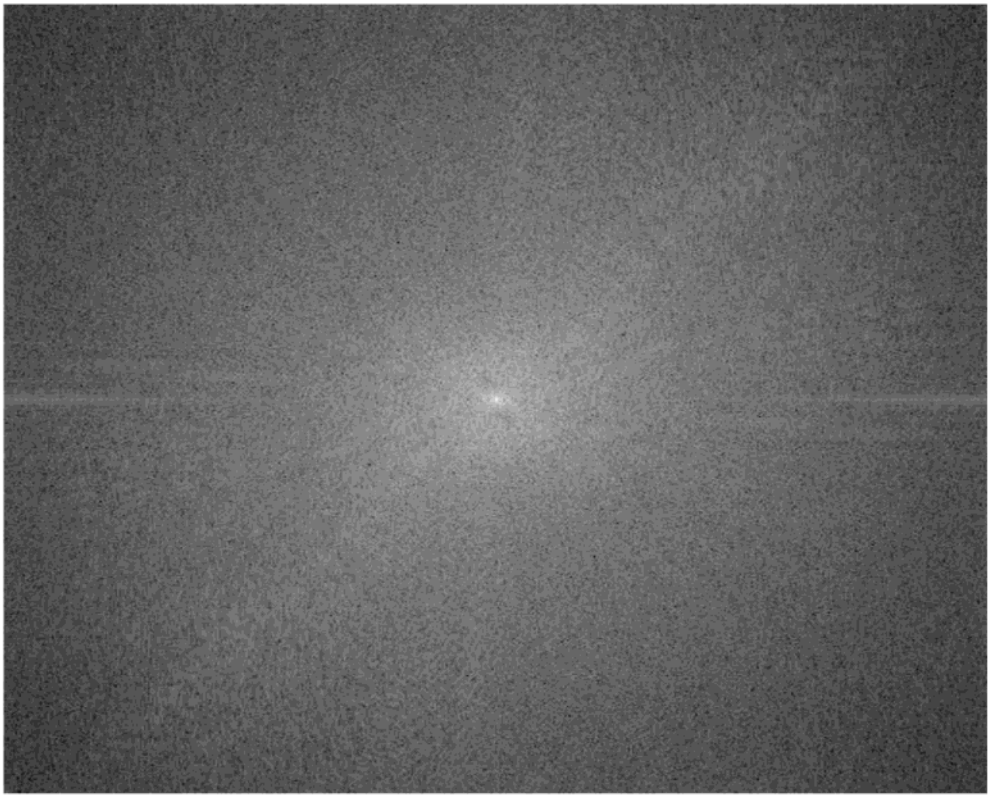
You can do the fourier transform first of X, then over Y and vise versa

36



Slide: Freeman & Durand

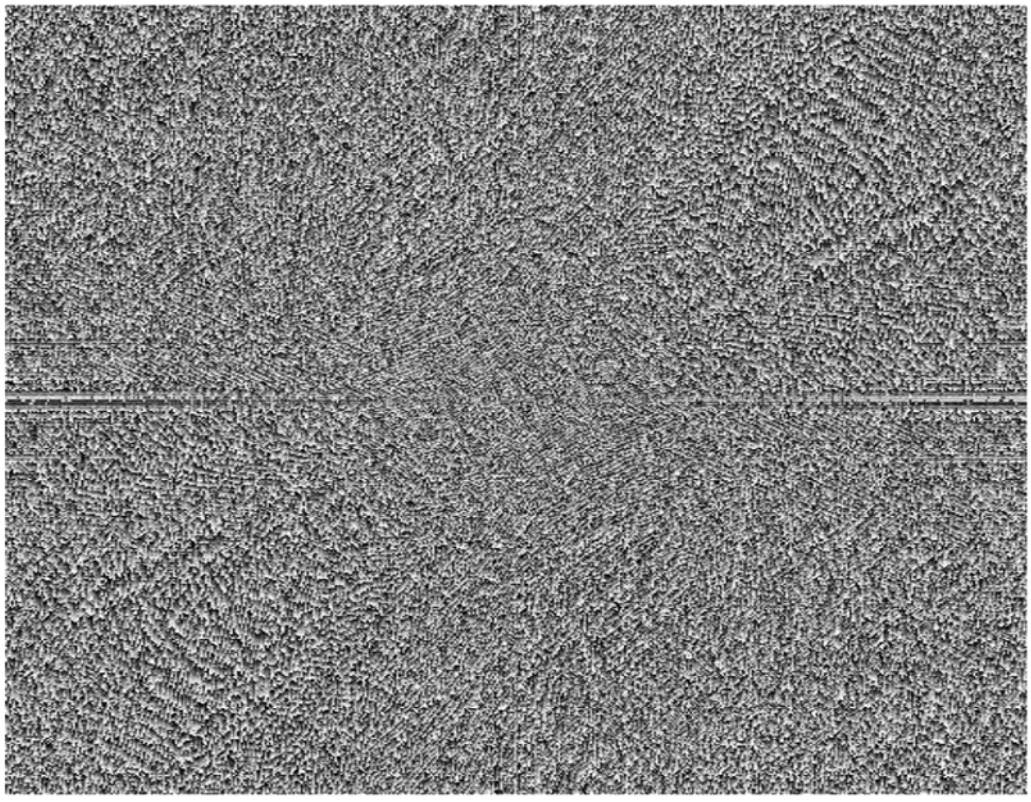
37



This is the magnitude transform of the cheetah pic

Slide: Freeman & Durand

38



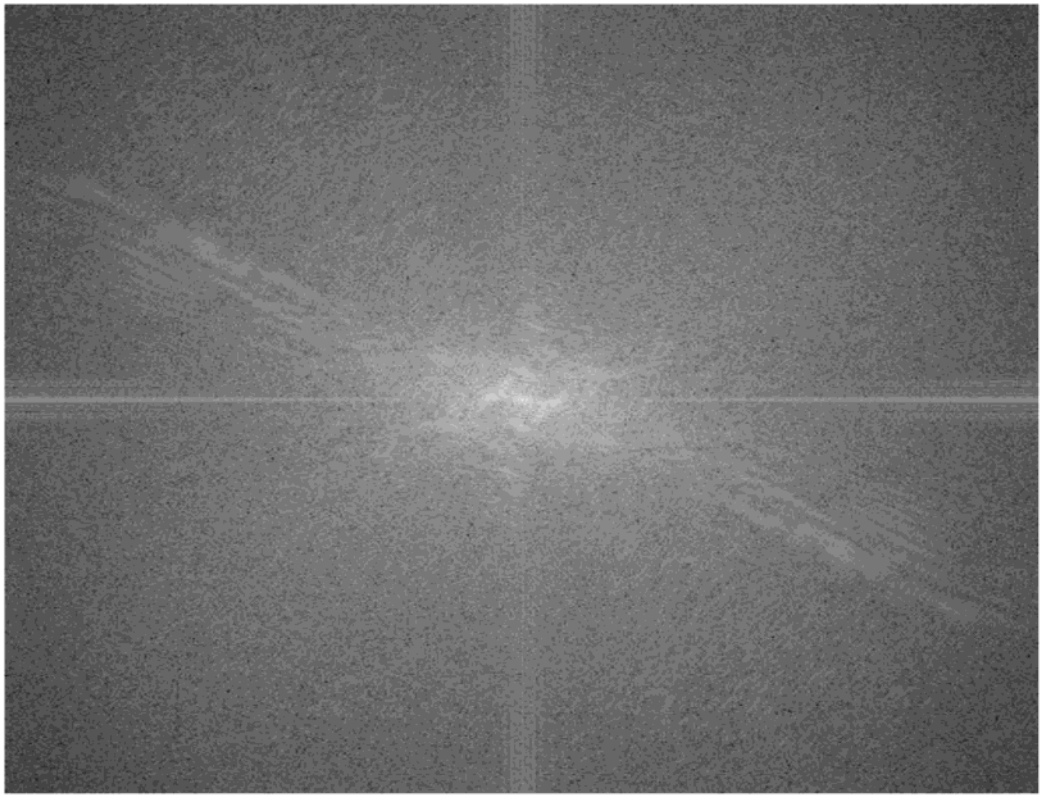
This is the phase transform of the cheetah pic

Slide: Freeman & Durand



Slide: Freeman & Durand

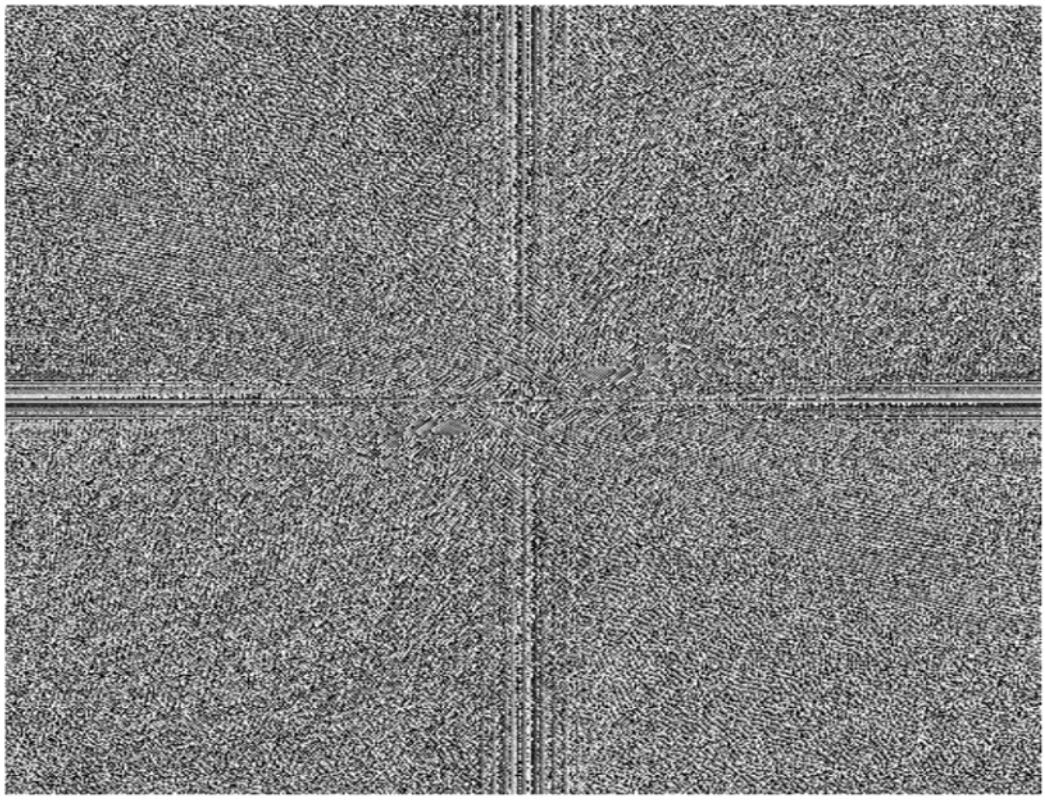
40



This is the magnitude transform of the zebra pic

Slide: Freeman & Durand

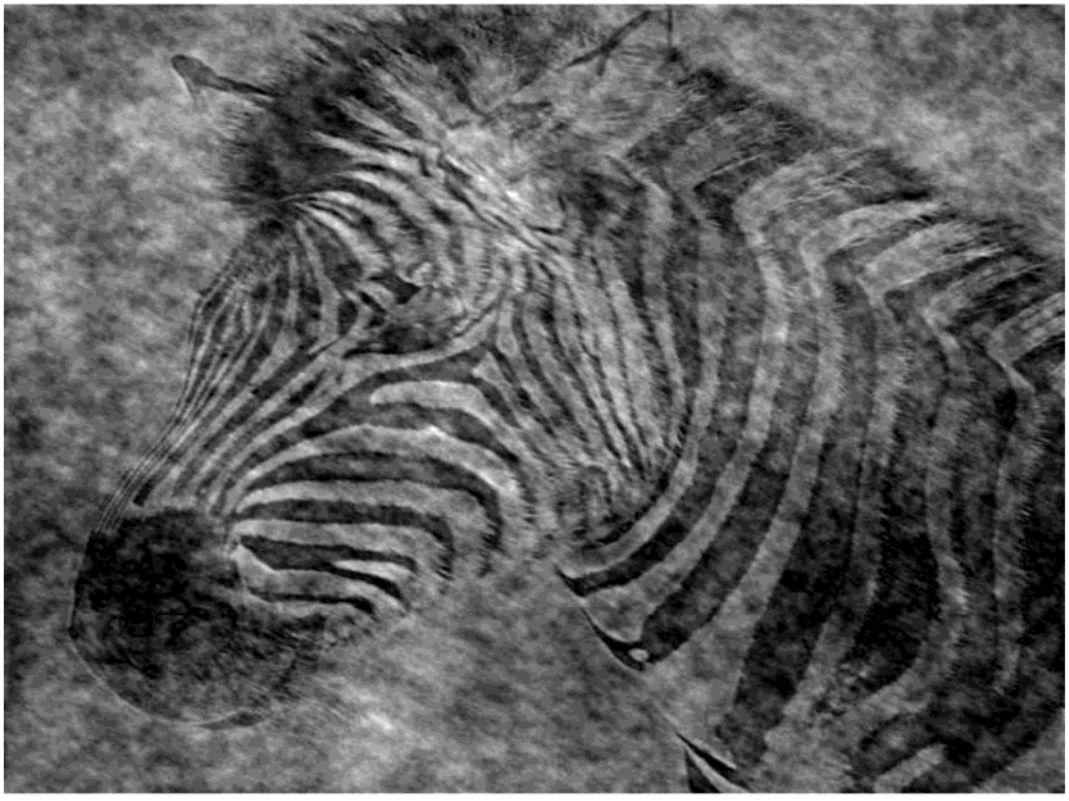
41



This is the phase transform of the zebra pic

Slide: Freeman & Durand

42



Reconstruction with zebra phase, cheetah magnitude

Slide: Freeman & Durand



Reconstruction with cheetah phase, zebra magnitude

Slide: Freeman & Durand

Fourier Properties

1. **Separability of the 2D transform:**

   2D Transform = *successive applications of 1D transforms*

2. **Linearity:** $f+g \leftrightarrow F+G$

   $af \leftrightarrow aF$

3. **Image average:** $F(0,0) = avg(f(x,y))$

4. **Shift ⇔ phase-change:** $g(x)=f(x+a) \leftrightarrow G(u)=F(u)e^{2\pi iua}$ *(continuous)*

   $G(u)=F(u)e^{2\pi iua/N}$ *(discrete)*

   $|G(u)| = |F(u)|$

* *Same magnitude; only phase shift.*
* *Can be used to recover shift (a,b) between two images f(x,y) & g(x,y):*

$G(u,v)/F(u,v)=e^{2\pi i(\frac{ua}{N}+\frac{vb}{M})}$ $\xrightarrow{FFT^{-1}}$ $\delta(x+a,y+b)$

Fourier Properties

5. **Scaling/Shrinking:** $f(ax) \leftrightarrow \frac{1}{|a|} F\left(\frac{u}{a}\right)$

6. **Derivative:** $g(x)=\frac{d}{dx} f(x) \leftrightarrow G(u)= 2\pi i\, u\, F(u)$
   * $G(0)=0$.
   * *Taking the derivative increases high frequencies (noise, edges)*
   * *Behaves like a high-pass filter (HPF).*

7. **Rotation:** *rotation of* $f(x,y)$ *by* θ $\leftrightarrow$ *rotation of* $F(u,v)$ *by* θ
   (can be shown using polar coordinates)

8. **Principle of Uncertainty:** No meaning to "*a frequency at a point*" !
   - *every value* $f(x)$ *contributes to all frequencies* $F(u)$
   - *every frequency* $F(u)$ *contributes to all values* $f(x)$

Fast Fourier Transform - FFT

$$F(u) = \frac{1}{N}\sum_{x=0}^{N-1} f(x) e^{\frac{-2\pi i u x}{N}} \qquad u = 0, 1, 2, ..., N-1$$

$O(N^2)$ operations, if performed as is

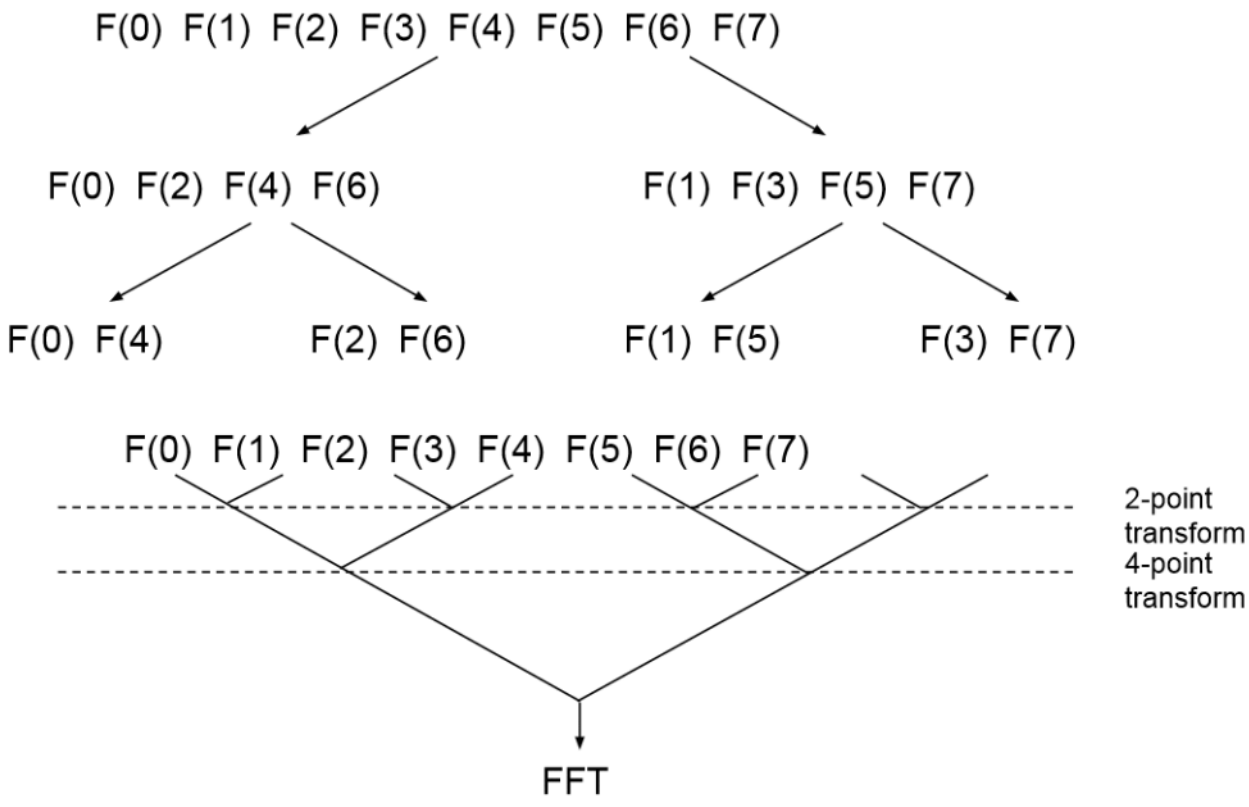
FFT

$$F(u) = \underbrace{\frac{1}{N}\sum_{x=0}^{N/2-1} f(2x) e^{\frac{-2\pi i u 2x}{N}}}_{\text{even } x} + \underbrace{\frac{1}{N}\sum_{x=0}^{N/2-1} f(2x+1) e^{\frac{-2\pi i u (2x+1)}{N}}}_{\text{odd } x}$$

$$= \frac{1}{2}\left[\underbrace{\frac{1}{N/2}\sum_{x=0}^{N/2-1} f(2x) e^{\frac{-2\pi i u x}{N/2}}}_{\text{Fourier Transform of of N/2 even points}} + e^{\frac{-2\pi i u}{N}} \underbrace{\frac{1}{N/2}\sum_{x=0}^{N/2-1} f(2x+1) e^{\frac{-2\pi i u x}{N/2}}}_{\text{Fourier Transform of of N/2 odd points}} \right]$$

The Fourier transform of N inputs, can be performed as 2 Fourier Transforms of N/2 inputs each + one complex multiplication and addition for each value. Thus, if F(N) is the computation complexity of FFT:

$$F(N)=F(N/2)+F(N/2)+O(N) \Rightarrow F(N)=N \log N$$



FFT : O(N log(N)) operations

FFT of NxN Image: $O(N^2 \log(N))$ operations